

Life unknown: On science and research freedom

The **cosmos** holds limitless mysteries for the **constrained** human mind

The University of Cambridge issued a press release regarding a paper co-authored by one of its scientists that reported an **exoplanet** called K2-18b may be **habitable**, with the **misleading** headline, Strongest **hints** yet of biological activity outside the solar system. **The paper**, published on April 17, 2025, **made headlines** because **astronomers** have been interested for some time in this exoplanet's **peculiar** surface physiochemistry. The paper is **to its credit** **sedate** in its **prognostications**: the authors have reported detecting the presence of dimethyl sulphide or dimethyl disulphide in K2-18b's atmosphere; the former is considered a sign of life on the earth and could be on the exoplanet. Independent scientists and the paper's authors have **called for dispassionate** efforts to refine the data and reduce uncertainties as well as to **double-check** the **implications**. If **the history** of the search for **biomarkers** on other planets both within and without the solar system **is** any guide, there is no guarantee that the **findings** will not be **overturned**. The cosmos still **harbours** too many mysteries for us to not be surprised by new information awaiting discovery. **Research** of this nature — initiated by curiosity and a **pursuit** of the unknown — **has** sometimes been **overtaken** by **hype**. Unfortunately, it faces another threat today: **funding** cuts of the type U.S. President Donald Trump **has unleashed** in his country.

After **downsizing** or cancelling grants issued via the National Institutes of Health, Mr. Trump began holding entire universities **hostage**: **drawdown** DEI initiatives and **rein in** student protests or **forgo** federal funding, he seemed to say. The **institutions** affected **count** some of the world's best in their ranks and the funding **withheld**, a **non-trivial fraction** of that spent on certain subjects worldwide. Earlier this month, the White House's budget **proposal** for the 2026 **fiscal** **revealed** its plans to cut National Oceanic and Atmospheric Administration funding by 27%, **nix** most spending for science at NASA's Goddard Space Flight Center, and **halve** NASA's earth-science budget. The essential programme is to control the practice and pursuit of science that does not **align with** Republican state ideology. Curiosity-driven **research**, such as that **concerned** with the possibility of extra-terrestrial life, both **proves** universities' freedom to explore the unknown and fulfils the **innate** curiosity of **humankind**. However, the Trump government's policies leave whatever research freedom that remains very precious yet also more **vulnerable** to **misinformation** and ideological capture, and the world itself more **short-sighted**.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.
- **Nix** (verb) – put an end to; cancel. रद्द करना।
- **Concern** (with) (verb) – relate to; be about. से संबंधित

Vocabulary

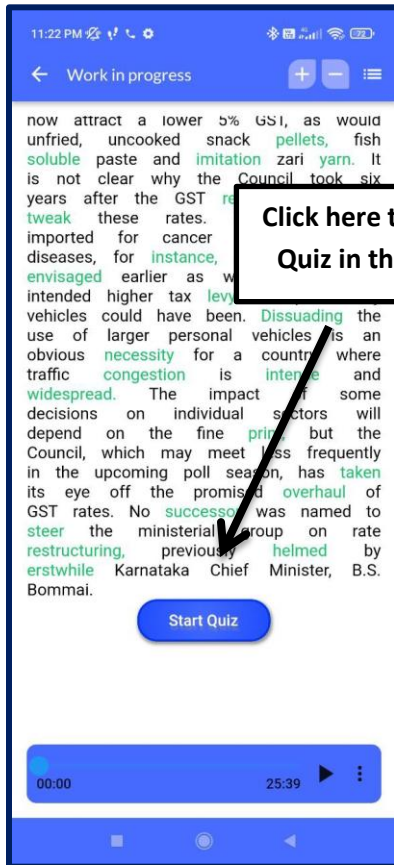
1. **Cosmos** (noun) – the universe seen as a well-ordered whole. ब्रह्मांड
2. **Constrained** (adjective) – restricted, confined, limited, restrained, inhibited सीमित
3. **Exoplanet** (noun) – a planet that orbits a star outside the solar system. सौरमंडल के बाहर का ग्रह
4. **Habitable** (adjective) – livable, inhabitable, survivable, tenantable, suitable रहने योग्य
5. **Misleading** (adjective) – deceptive, confusing, false, delusive, erroneous भ्रामक
6. **Hint** (verb) – suggest, imply, indicate, allude, insinuate संकेत देना
7. **Make headlines** (phrase) – attract attention, become news, gain publicity, become prominent, hit the news सुर्खियाँ बनाना
8. **Astronomer** (noun) – a person who studies astronomy खगोलशास्त्री
9. **Peculiar** (adjective) – strange, unusual, odd, abnormal, distinctive विचित्र
10. **To its credit** (phrase) – commendably, admirably, rightfully, appreciably, notably प्रशंसा की बात यह है कि
11. **Prognostication** (noun) – prediction, forecast, prophecy, projection, foretelling पूर्वानुमान
12. **Call for** (phrasal verb) – demand, request, require, urge, seek की मांग करना
13. **Dispassionate** (adjective) – impartial, unemotional, detached, objective, neutral निष्पक्ष
14. **Double-check** (verb) – verify, recheck, confirm, review, reassess दोबारा जांचना
15. **Implication** (noun) – consequence, effect, inference, indication, outcome निहितार्थ / प्रभाव
16. **Biomarker** (noun) – a measurable indicator of some biological state or condition. जैविक सूचक
17. **Finding** (noun) – result, conclusion, discovery, observation, outcome निष्कर्ष / खोज
18. **Overtake** (verb) – reverse, annul, invalidate, revoke, cancel पलटना / रद्द करना
19. **Harbour** (verb) – contain, shelter, possess, conceal, foster छिपाना / रखना
20. **Pursuit** (noun) – quest, search, chase, aim, striving खोज / प्रयास
21. **Overtake** (verb) – surpass, exceed, overwhelm, eclipse, outstrip आगे निकल जाना
22. **Hype** (noun) – publicity, promotion, sensation, exaggeration, buzz बहुत प्रचार

23. **Unleash** (verb) – release, launch, trigger, initiate, let loose आरंभ करना
24. **Downsize** (verb) – reduce, shrink, trim, scale down, cut back आकार घटाना
25. **Hostage** (noun) – captive, pawn, detainee, prisoner, leverage बंधक
26. **Drawdown** (verb) – reduce, diminish, scale down, lessen, cut घटाना
27. **Rein in** (phrasal verb) – control, restrain, suppress, limit, curb नियंत्रण करना
28. **Forgo** (verb) – give up, relinquish, waive, abandon, sacrifice त्यागना
29. **Withhold** (verb) – hold back, retain, suppress, refuse, deny रोकना
30. **Non-trivial** (adjective) – significant, important, considerable, notable, meaningful महत्वपूर्ण
31. **Fraction** (noun) – portion, segment, piece, part, division अंश
32. **Fiscal** (noun) – financial year वित्तीय वर्ष
33. **Halve** (verb) – divide, split, reduce, bisect, diminish आधा करना
34. **Align** (with) (verb) – match, coordinate, harmonize, conform, adjust मेल खाना
35. **Innate** (adjective) – inborn, natural, inherent, intrinsic, instinctive जन्मजात
36. **Humankind** (noun) – humanity, mankind, people, humans, species मानव जाति
37. **Vulnerable** (to) (adjective) – susceptible, exposed, defenseless, sensitive, unprotected असुरक्षित
38. **Misinformation** (noun) – falsehood, disinformation, untruth, fabrication, lie गलत सूचना
39. **Short-sighted** (adjective) – narrow-minded, unwise, imprudent, myopic, thoughtless अल्पदृष्टि

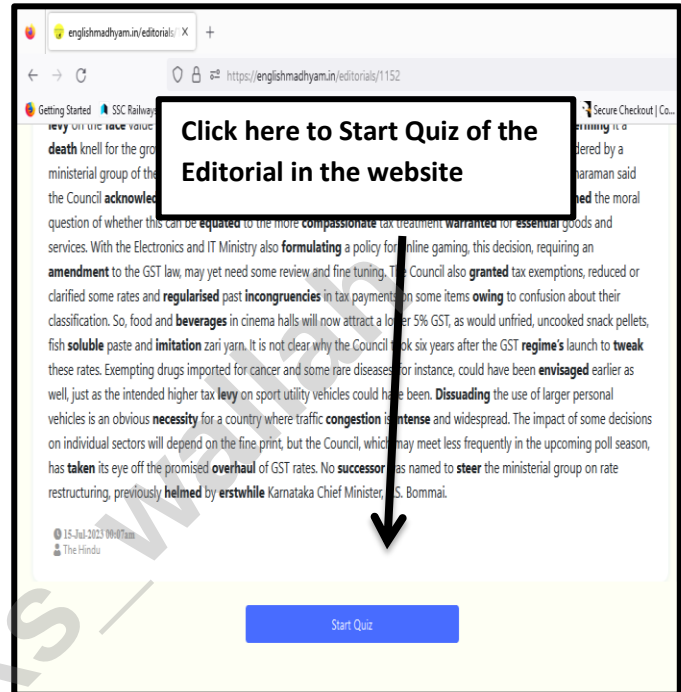
Summary of the Editorial

1. **Headline Misrepresentation** – A University of Cambridge press release gave a misleading headline about possible life on exoplanet K2-18b, exaggerating the findings.
2. **Actual Findings** – The scientific paper reported the possible presence of dimethyl sulphide, a compound linked to biological activity on Earth, in the atmosphere of K2-18b.
3. **Scientific Caution** – The paper itself was cautious and not definitive; the authors urged careful interpretation and further validation of the results.
4. **Uncertainty in Space Research** – History shows that discoveries about extraterrestrial life often face revisions, reflecting the complexity of cosmic research.
5. **Value of Mystery** – The editorial emphasizes the vast unknowns in the cosmos and the importance of maintaining openness to new discoveries.
6. **Role of Curiosity** – Research into extraterrestrial life is driven by human curiosity and the desire to explore the unknown.
7. **Threat of Hype** – Sometimes, the scientific search is compromised by media sensationalism and public hype, diverting focus from actual findings.
8. **Political Threats to Science** – The Trump administration is accused of undermining scientific freedom by using funding as a political weapon.
9. **Cuts to Major Science Funding** – Trump's policies have led to reduced grants via the NIH and threatened funding to top U.S. universities.
10. **Conditional Funding** – Institutions were pressured to scale back on DEI (Diversity, Equity, Inclusion) efforts and student protests to retain federal support.
11. **Budget Cuts in 2026 Proposal** – The 2026 U.S. budget plan includes deep cuts to NOAA (27%), and reductions to NASA's Goddard and Earth-science programmes.
12. **Ideological Control** – These budget decisions reflect an intent to shape science policy according to Republican ideologies, limiting academic freedom.
13. **Impact on Global Research** – U.S. funding cuts have global implications, as affected institutions play a major role in worldwide scientific research.
14. **Risk to Curiosity-Driven Research** – Studies like the search for alien life, which exemplify the freedom to explore, are now more vulnerable to external pressures.
15. **Call for Protection of Science** – The editorial concludes with a warning: without defending scientific independence, the world risks becoming intellectually impoverished and misinformed.

Practice Exercise: Banking Pattern Based



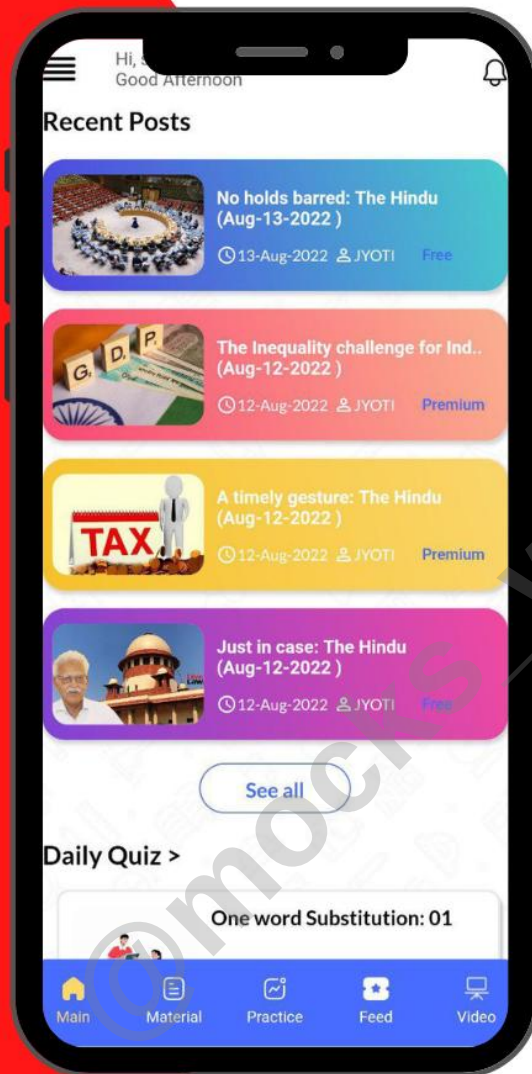
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